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Voltage Follower, Part III

Welcome to the 35th article in the "Circuit Intuitions" column series. As the title suggests, each article provides insights and intuitions into circuit design and analysis. These articles are aimed at undergraduate students but may serve the interests of other readers as well. If you read this article, I would appreciate your comments and feedback as well as your requests and suggestions for future articles in this series. Please e-mail me your comments at ali@ece.utoronto.ca.

In the previous two articles [1], [2] in this series, we reviewed the general characteristics of a simple voltage follower (SVF), as shown in Figure 1(a), and compared them against those of a flipped voltage follower (FVF) [3], [4], as shown in Figure 1(b). An SVF exhibits an infinite input resistance, a small output resistance, and a voltage gain of slightly less than 1 V/V. In contrast, an FVF exhibits a much-reduced output resistance (by a factor of $g_m r_o$) while maintaining the same voltage gain as that of an SVF. In this article, we look at another variation of the FVF, known as the *super source follower* (*SSF*) [5], before making detailed comparisons among the three designs and providing some intuitions.

In this article, we look at another variation of the FVF, known as the super source follower, before making detailed comparisons among the three designs and providing some intuitions.

Figure 1(c) shows a schematic of an SSF, which consists of an input transistor M_1 with a bias current of I_{B1} and an auxiliary PMOS transistor M_2 with a bias current of I_{B2}. For simplicity, we assume both transistors to have the same bias currents, $I_{B1} = I_{B2} = I_B$; the same short circuit transconductance, $g_{m1} = g_{m2} = g_m$; and the same output resistance, $r_{o1} = r_{o2} = r_o$. That is, we assume that the bottom current source provides twice the current of the top current source. Further, we assume both transistors to be in the saturation region. Let us now find an expression for the output voltage of this circuit when we apply a low-frequency small-signal voltage to the input. In general, the voltage of any node in a linear time-invariant circuit can be found by multiplying its short circuit current to the ground and its output resistance [6]. To find the short circuit current, we use the simplified small-signal circuit of Figure 2(a), where all of the dc voltage and current sources are zeroed. The short circuit current i_{sc} is simply the sum of i_1 and i_2. However, i_1 is zero because it leads to an open circuit at the gate of M_2. The voltage at the drain of M_1 is simply the product of its short circuit current to the ground ($-g_m v_{in}$) and its equivalent resistance (r_o). Therefore, we have a voltage of $-g_m r_o v_{in}$ at the gate of M_2; when we multiply this by its g_m, we find

$$i_{sc} = i_2 = (g_m r_o) \times g_m v_{in}$$

where, for simplicity, we have assumed the two transistors to have the same g_m. This equation is, in fact, identical to what we found in the previous article in this series [2] for the FVF. That is, the short circuit currents of both the FVF and the SSF are $g_m r_o$ times larger than the short circuit current of the SVF.

Let us now find the output resistance of the SSF with the help of the equivalent circuit shown in Figure 2(b). To find the output resistance, we ground the input node, apply an ideal signal source v_x to the output node, and measure the resulting signal current i_x. Again, for the same reasons provided earlier, $i_x = i_1 + i_2$, with $i_1 = 0$. We observe that $i_2 = g_m \times v_{d1} + v_x / r_o$. We find v_{d1} using the equivalent circuit of Figure 2(c) by multiplying the short circuit current at the drain of M_1, $i_{sc1} = v_x(g_{me} + 1/r_o)$ and r_o, where $g_{me} = g_m + g_{mb}$ is the effective short circuit transconductance of the transistor, including the body effect. In other words, $v_{d1} = (1 + g_{me} r_o) \times v_x$; hence,

$$i_x = v_x g_m (1 + g_{me} r_o) + \frac{v_x}{r_o}.$$

If we assume $g_m r_o \gg 1$, only the first term in i_x will be dominant, and we can write

$$R_{out} = \frac{v_x}{i_x} \approx \left(\frac{1}{g_m r_o}\right)\left(\frac{1}{g_{me}}\right).$$

Again, this equation is identical to what we found for the FVF. In both cases, the output resistance is reduced by a factor of $g_m r_o$ compared to the SVF. Recall that this is the same factor by which the short circuit current

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increased for both designs. Therefore, the output voltage, which is the product of the two, will remain the same for all three designs:

$$\frac{V_{\text{out}}}{V_{\text{in}}} \approx \frac{g_m}{g_{me}}.$$

At this point, a question may be what we have achieved in moving from the SVF to the FVF and the SSF. While the output voltages of all three designs closely follow their respective input voltages, the FVF and the SSF have much lower output resistances compared to that of the SVF. This means the FVF and the SSF are closer to being ideal voltage followers with zero output resistance.

Now, let us take a closer look at the three designs, as shown in Figure 1, to see intuitively why the FVF and the SSF designs are superior to the SVF design. For this intuition, we totally ignore the body effect in all three designs, and we further assume an infinite r_o for all transistors. Let us start from the SVF, where transistor M_1 is providing both the voltage and the current to the load. We show that a single transistor cannot be good at both. Under no load, the current through M_1 will be fixed at I_B, implying a fixed v_{GS}. This is a direct consequence of M_1 being in the saturation region, where

$$i_{\text{DS}} = \frac{\mu_n C_{OX}}{2} \times \frac{W}{L} \times (v_{GS} - V_T)^2$$

While the output voltages of all three designs closely follow their respective input voltages, the FVF and the SSF have much lower output resistances compared to that of the SVF.

Given that $v_{\text{OUT}} = v_{\text{IN}} - v_{GS}$ we can safely conclude that the output voltage is simply a shifted-down version of the input voltage, where the dc shift (v_{GS}) is independent of the signal. In other words, for small signals, $v_{\text{out}} = v_{\text{in}}$. How about when there is a nonzero current signal going to the load? In this case, the current through M_1 is no longer constant but is the sum of the dc bias current and the load current. This varying i_{DS} implies a varying v_{GS}. That is, the level shift from input to output is no longer constant but changes with the signal. This destroys the perfect voltage-following operation that the SVF enjoys

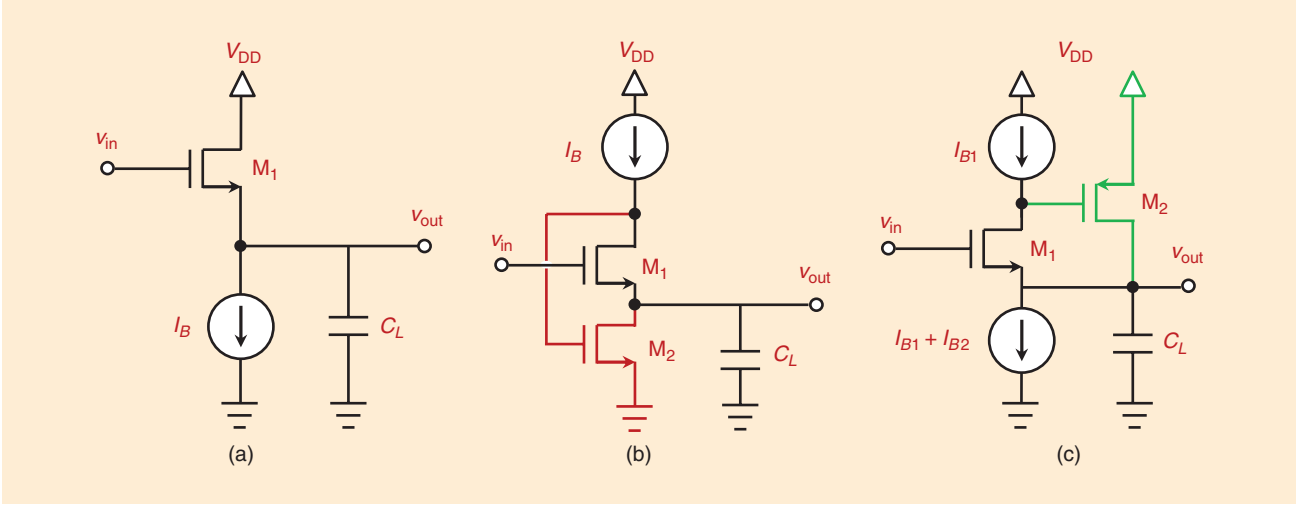


FIGURE 1: A schematic of (a) an SVF, (b) a flipped voltage follower, and (c) a super source follower.

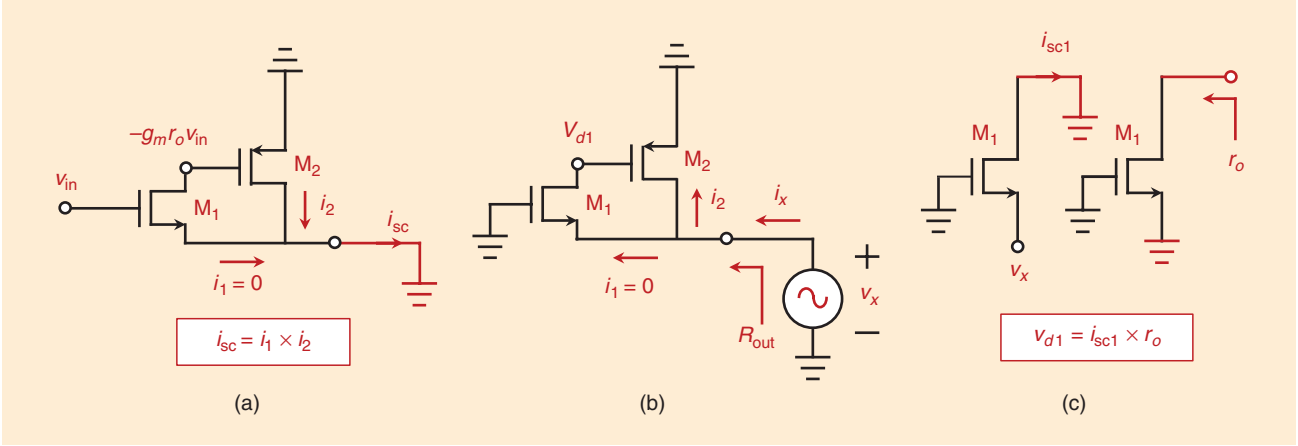


FIGURE 2: A small-signal equivalent circuit for the SSF (a) to find its short circuit transconductance, (b) to find its output resistance, and (c) to find the drain voltage of M_1 in (b).

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TABLE 1. NMOS AND PMOS TRANSISTOR PARAMETERS USED IN SIMULATIONS OF THE SVF, FVF, AND SSF.

PARAMETER	M_1 (NMOS)	M_2 (PMOS)
W/L	400 nm/45 nm	1150 nm/45 nm
$\|V_T\|$	0.43 V	0.27 V
g_m	0.5 mS	0.5 mS
g_{mb}	0.15 mS	0.15 mS
r_o	10.5 kΩ	9.3 kΩ

when there is no load current. To summarize, when we draw no signal current from M_1, M_1 acts as a perfect follower. But what is the point of a follower when it cannot deliver current to the load? This was the whole point of designing a voltage follower. So, what can we do? The idea that is embedded in both the FVF and the SSF is to separate the task of voltage following and current delivery to two distinct transistors M_1 and M_2.

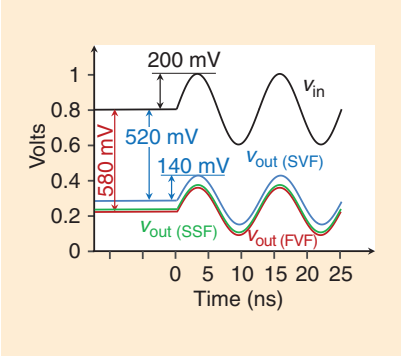


FIGURE 3: An input voltage waveform (in black) and its corresponding output voltage waveform for (a) an NMOS SVF (in blue), (b) an NMOS FVF (in red), and (c) an NMOS SSF (in green).

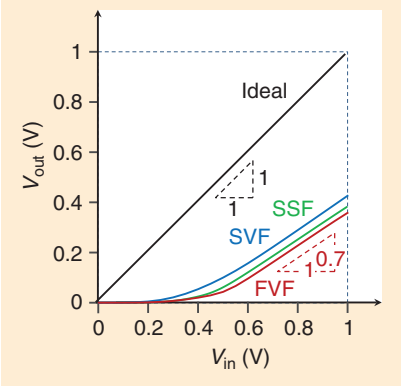


FIGURE 4: Simulated dc characteristics of the SVF (in blue), FVF (in red), and SSF (in green) versus that of an idea voltage follower (in black).

In the FVF, the current through M_1 is fixed by I_B. No signal current can go through M_1 simply because it is in series with an ideal dc current source. This would guarantee a fixed v_{GS1} and, hence, a perfect act of voltage following under load or no load. Transistor M_2, on the other hand, is fully responsible to provide the load current. The reader is encouraged at this point to pause and ponder how M_2 knows when to supply current to the load.

To answer this question, we must drop our infinite r_o assumption. Indeed, with an infinite r_o, the small-signal voltage at the gate of M_2 is not well defined, and M_2 will have no way of knowing how much current to provide to the load. With a finite r_o, however, we can intuitively explain this as the following. Any small increment at the output node that goes beyond the increment of the input node will create a corresponding voltage increment at the drain of M_1 (which is connected to the gate of M_2) and a corresponding increment in the current that M_2 sinks from the output node, causing a decrement in the output voltage. This is called *negative feedback*, and we will review this at length in a future article in this series. Here, we suffice to say that feedback is employed in both the FVF and the SSF to reduce their output resistance relative to that of the SVF.

The design of the SSF follows the exact same principle. As in the FVF, M_1 will carry zero signal current and, hence, delivers the input voltage to the load, while M_2 provides the current to the load. In summary, the two tasks of voltage delivery and current delivery to the load are now separated and given to M_1 and M_2, respectively.

We have used the parameters shown in Table 1 with a nominal bias current of $I_B = I_{B1} = I_{B2} = 50\ \mu$A and a load capacitance of $C_L = 100$ fF to simulate the SVF, the FVF, and the SSF circuits and to compare their dc characteristics and frequency responses. Figure 3 shows that, for the parameters used, all three designs shift down the input voltage waveform by around the same dc value but with a slightly higher shift in the case of the FVF and SSF: ~580 mV for the FVF and SSF versus ~520 mV for the SVF. This larger shift in the FVF and the SSF is due to a small voltage drop across the nonideal current source we have used in our simulations. Finally, note that, as expected, all three designs exhibit a similar voltage gain of g_m/g_{me}, which is around 0.7 V/V.

Figure 4 compares the dc characteristics of the three followers against that of an ideal voltage follower. We observe that the outputs of all three designs follow their respective inputs only for input voltages larger than about 0.6 V.

Finally, given a reduction in the output resistance of the FVF and SSF compared to that of the SVF, by a factor of $g_m r_o$, we expect the FVF and SSF to have an increased bandwidth by the same factor. This is, indeed, consistent with the simulation results shown in Figure 5, where both the FVF and SSF exhibit a similar bandwidth improvement of around five times over the SVF.

We conclude this article by offering two questions to explore:

- It appears that the SSF draws twice as much current as the FVF from the power supply, i.e., consuming twice the power while providing the same output resistance. Is there any advantage in using the SSF over the FVF?
- In general, under which circumstances do we use each of the SVF, the FVF, and the SSF in our designs?

(continued on p. 26)

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to ever-improving health and wellness for the human race.

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Appendix: Related Articles

[A1] D. A. Hall, K. A. A. Makinwa, and T. Jang, "Quantifying biomedical amplifier efficiency: The noise efficiency factor," *IEEE Solid-State Circuits Mag.*, vol. 15, no. 2, pp. 28–33, Spring 2023, doi: 10.1109/MSSC.2023.3256353.

[A2] G. L. Barbruni, D. Ghezzi, and S. Carrara, "Challenges for miniaturized neurostimulators: Design of wireless Bio/CMOS interfaces for brain stimulation," *IEEE Solid-State Circuits Mag.*, vol. 15, no. 2, pp. 34–40, Spring 2023, doi: 10.1109/MSSC.2023.3246004.

[A3] A. Emami and S. Sharma, "3D localization of miniaturized devices for precision medicine: Enabling wireless surgical navigation, health monitoring, and targeted drug delivery," *IEEE Solid-State Circuits Mag.*, vol. 15, no. 2, pp. 41–48, Spring 2023, doi: 10.1109/MSSC.2023.3247686.

[A4] J. Yoo, "Body area network: Connecting and powering things together around the human body," *IEEE Solid-State Circuits Mag.*, vol. 15, no. 2, pp. 49–58, Spring 2023, doi: 10.1109/MSSC.2023.3264663.

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CIRCUIT INTUITIONS *(continued from p. 16)*

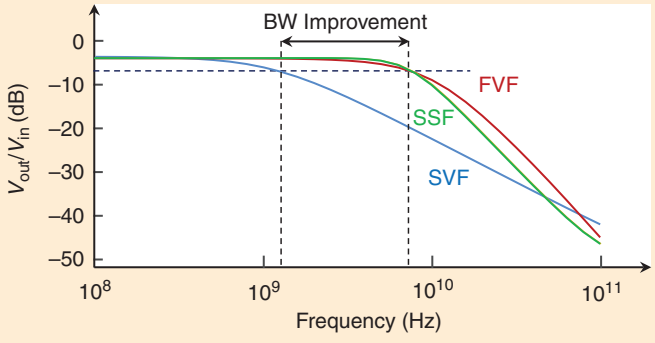


FIGURE 5: Frequency responses of the SVF (in blue), FVF (in red), and SSF (in green) showing the bandwidth (BW) improvement of the FVF and SSF over the SVF.

We will answer these questions and explore yet another voltage follower in the next article in this series.

In summary, we have shown that the SVF, the FVF, and the SSF all offer the same voltage gain of around 0.7 V/V. While the SVF burdens a single transistor to deliver both the voltage and the current to the load, the FVF and SSF divide these two tasks between two transistors, one delivering the voltage and one delivering the current. In doing so, the FVF and the SSF continue to follow the input voltage even when they need to supply varying signal currents to the load. All of this is to say that the FVF and the SSF have much lower output resistance compared to the SVF.

Acknowledgment

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